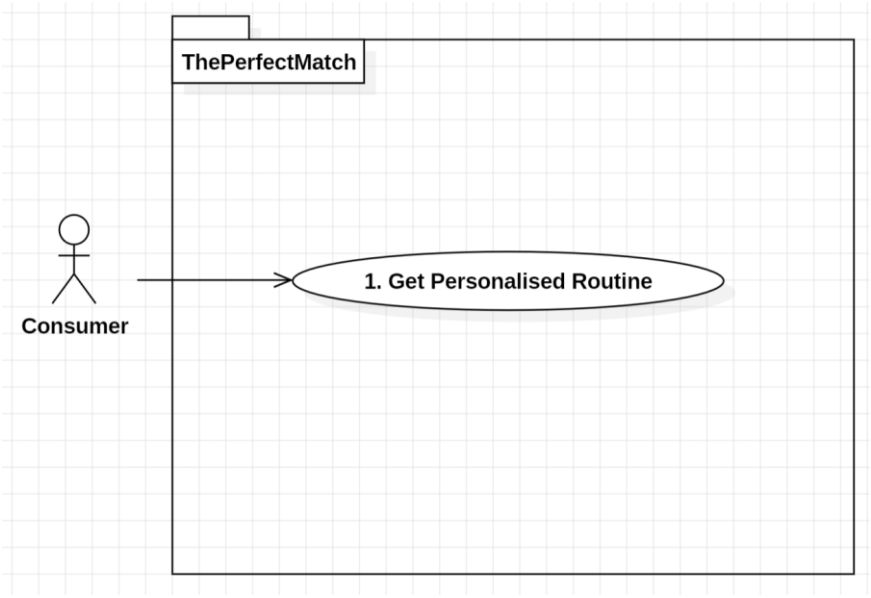
SYSTEM ANALYSIS

Contents

1. First Iteration 2
1.1 Use Case Diagram 2
1.2 Use Case Narratives for Iteration 1 2
1.2.1 Narrative for Use Case 1 – Get Personalised Routine 2
2. Second Iteration 5
2.1 Use Case Diagram 5
2.2 Use Case Narratives for Iteration 2 5
2.2.1 Narrative for Use Case 1 – Get Personalised Routine 5
2.2.2 Narrative for Use Case 2 – Get Expert Feedback 7
2.2.3 Narrative for Use Case 3 – AR Try-On 9
2.2.4 Narrative for Use Case 4 – Process Consumer Data 10
2.2.5 Narrative for Use Case 5 – Supply Product Data 11
3. Third Iteration 13
3.1 Use Case Diagram 13
3.2 Use Case Narratives for Iteration 3 13
3.2.1 Narrative for Use Case 1 – Get Personalised Routine 13
3.2.2 Narrative for Use Case 2 – Get Expert Feedback 13
3.2.3 Narrative for Use Case 3 – AR Try-On 13
3.2.4 Narrative for Use Case 4 – Process Consumer Data 13
3.2.5 Narrative for Use Case 5 – Supply Product Data 13
3.2.6 Narrative for Use Case 6 – Purchase Products 14
3.2.7 Narrative for Use Case 7 – Provide Skin Coaching 15
3.2.8 Narrative for Use Case 8 – Check Ingredient Safety 16
3.2.9 Narrative for Use Case 9 – Compare Products 17
3.2.10 Narrative for Use Case 10 – Create Community Content 18
4. Analysis Classes 20
4.1 Pair: Get Personalised Routine & Supply Product Data 20
4.2 Pair: Get Expert Feedback & Process Consumer Data 20
5. Sequence Diagrams 21
5.1 Pair: Get Personalised Routine & Supply Product Data 21
5.2 Pair: Get Expert Feedback & Process Consumer Data 21
6. References 22

1. First Iteration

1.1 Use Case Diagram



1.2 Use Case Narratives for Iteration 1

1.2.1 Narrative for Use Case 1 – Get Personalised Routine

USE CASE	Use Case: 1	Get Personalised Routine
Brief Description of Goal in Context	The Consumer wants to receive a personalised self-care routine, covering skincare, haircare, makeup or grooming. The system analyses the Consumer's inputs such as selfies, quiz answers, or smart sensor readings and produces a tailored routine that fits their profile, preferences, and goals.	
Preconditions	- The Consumer has installed ThePerfectMatch app and created an account. - The Consumer has provided necessary personal data (e.g. skin type, allergies, goals, preferences)	
Post Conditions, Success End Condition	- A personalised routine is generated and displayed to the Consumer. - Consumer can save, share, or view product suggeestions related to the routine.	
Long DESCRIPTION of Scenario	The Consumer opens ThePerfectMatch app and selects the option to get a personalized routine. The system requests input such as a selfie, answers to a few lifestyle or skin-related questions, or optional sensor data. It analyses the information received using AI algorithms and a product database. Based on the Consumer's profile and preferences, the system created a customised routine that includes recommended products and step-by-step guidance. The Consumer views the routine and may save or share it for later use.	

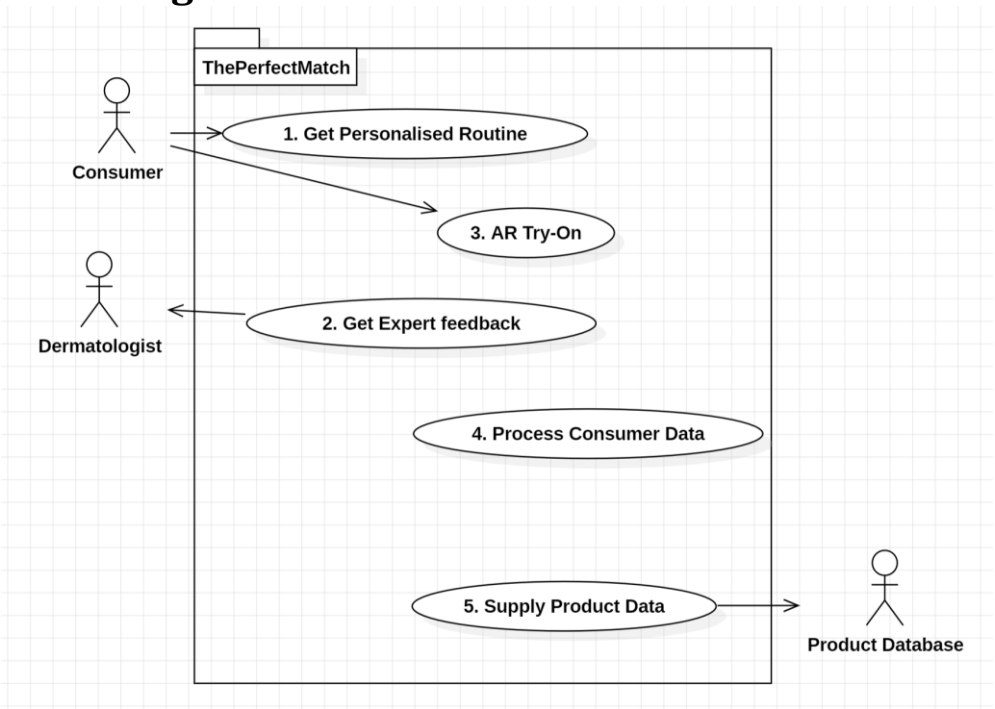
Main Flow		
Step	**Action**	**Alternate**
n.1.1	The Consumer opens ThePerfectMatch app.	
n.1.2	The system displays the main dashboard.	
n.1.3	The Consumer selects “Get Personalised Routine”.	
n.1.4	The system presents available input options (selfie, quiz, sensor).	
n.1.5	The Consumer chooses one input method.	
n.1.6	The system activates the selected input preference.	
n.1.7	The Consumer provides required data (image, answers, or readings).	E.F. 1.7
n.1.8	The system validates the received data.	
n.1.9	The system analyses the data using AI algorithms.	
n.1.10	The system retrieves relevant products from the product database.	E.F. 1.10
n.1.11	The system matches products to Consumer preferences and restrictions.	
n.1.12	The system generates a complete personalised routine.	
n.1.13	The system displays the routine with step-by-step guidance.	
n.1.14	The Consumer reveiws the routine.	
n.1.15	The Consumer saves the personalised routine for future use.	
n.1.16	End Use Case.	
EXCEPTIONS or ERROR Flow Description		
Step	**Branching Action** < Exception number m of Use Case n>	**Alternate**
E.F. 1.7.1	Selfie upload fails.	
E.F. 1.7.2	Consumer retries.	
E.F. 1.7.3	Consumer chooses another option.	
E.F. 1.7.4	Go to Main Flow 1.6	
E.F. 1.10.1	No product found.	
E.F. 1.10.2	System asks to broaden filters.	
E.F. 1.10.3	Go to Main Flow 1.7	
ALTERNATIVE or VARIATION Flow Description		
Step	**Branching Action** No alternative flows as no options given in Use Case.	**Alternate**

RELATED INFORMATION	Use Case: 1	Get Personalised Routine
Priority:	High – most important features of the app.	
Product: Performance	AI routine generation must complete within 5 seconds. [1].	
Product: Efficiency	The app should not be too large (under 150MB) [2]. It should not drain battery too quickly.	
Product: Reliability	The app should be available almost all the time (99.9% uptime) [3]. It can have a down time of 1 hour per month.	

External: Legislation	All Consumer data (selfies, health data) must comply with GDPR [4].
External: Ethical	The app must show a disclaimer, e.g. “Do not consider this app as a doctor.” It should avoid algorithmic bias.
Channels to actors	Consumer interacts through mobile app UI. Sensors connect via Bluetooth.
OPEN ISSUES	1. Source of verified product data. 2. Integration of smart wearables.
Due Date	Dec 2025
…any other management information…	

2. Second Iteration

2.1 Use Case Diagram



2.2 Use Case Narratives for Iteration 2

2.2.1 Narrative for Use Case 1 – Get Personalised Routine

USE CASE	Use Case: 1	Get Personalised Routine
Brief Description of Goal in Context	The Consumer wants to receive a personalised self-care routine that now includes improved AI-driven recommendations and verified product information. The system analyses Consumer inputs (selfies, quiz answers, or smart sensor readings), retrieves verified product data from the Product Database, and produces a tailored routine that fits the Consumer's profile, preferences, and goals.	
Preconditions	- The Consumer has installed ThePerfectMatch app and created an account. - The Consumer has provided necessary personal data (e.g. skin type, allergies, goals, preferences). - AI Engine and Product Database connections are active.	
Post Conditions, Success End Condition	- A personalised routine is generated and displayed to the Consumer. - The Consumer can save, share, or try products virtually. - The structured routine is stored for possible dermatologist feedback.	
Long DESCRIPTION of Scenario	The Consumer opens ThePerfectMatch app and selects the option to get a personalized routine. The system requests input such as a selfie, answers to a few lifestyle or skin-related questions, or optional sensor data.	

	The AI Engine processes this input, identifies relevant features (skin tone, texture, dryness, etc.), and retrieves verified product data from the Product Database. Based on the combined data, the system generates a fully personalised self-care routine including recommended products and usage instructions. The routine is saved in the Consumer's profile, and results can now be shared for professional review in future consultations.

Main Flow

Step	Action	Alternate
n.1.1	The Consumer opens ThePerfectMatch app.	
n.1.2	The system displays the main dashboard.	
n.1.3	The Consumer selects "Get Personalised Routine".	
n.1.4	The system presents available input options (selfie, quiz, sensor).	
n.1.5	The Consumer chooses one input method.	
n.1.6	The system activates the selected input preference.	
n.1.7	The Consumer provides required data (image, answers, or readings).	E.F. 1.7
n.1.8	The system validates the received data.	
n.1.9	The AI Engine analyses the data using trained models.	
n.1.10	The AI Engine retrieves relevant products from the Product Database.	E.F. 1.10
n.1.11	The system matches products to Consumer preferences and restrictions.	
n.1.12	The system generates a complete personalised routine.	
n.1.13	The system displays the routine with step-by-step guidance.	
n.1.14	The Consumer reveiws the routine.	
n.1.15	The Consumer saves the personalised routine for future use.	
n.1.16	End Use Case.	

EXCEPTIONS or ERROR Flow Description

Step	Branching Action < Exception number m of Use Case n>	Alternate
E.F. 1.7.1	Selfie upload fails.	
E.F. 1.7.2	Consumer retries.	
E.F. 1.7.3	Consumer chooses another option.	
E.F. 1.7.4	Go to Main Flow 1.6	
E.F. 1.10.1	No product found.	
E.F. 1.10.2	System asks to broaden filters.	
E.F. 1.10.3	Go to Main Flow 1.7	

ALTERNATIVE or VARIATION Flow Description

Step	Branching Action No alternative flows as no options given in Use Case.	Alternate

RELATED INFORMATION	Use Case: 1	Get Personalised Routine
Priority:	High – most important features of the app.	

Product: Performance	AI routine generation must complete within 5 seconds. [1].
Product: Efficiency	The app should support offline mode for previously saved routines.
External: Ethical	Improved bias testing on diverse datasets.
Channels to actors	Consumer interacts through mobile app UI. Sensors connect via Bluetooth. AI Engine and Product Database communicate through internal APIs.
OPEN ISSUES	1. Improve product reccomendation accuracy through Consumer feedback. 2. Improve personalisation for specific conditions. 3. Implement office caching for requent and prevesouly saved routines.
Due Date	Dec 2025
...any other management information...	

2.2.2 Narrative for Use Case 2 – Get Expert Feedback

USE CASE	Use Case: 2	Get Expert Feedback
Brief Description of Goal in Context	The dermatologist provides expert feedback on AI-generated Consumer routines. The system securely sends Consumer data (images, routines, quiz data) to the dermatologist, who analyses and provides professional recommendations or adjustments to the Consumer's existing routine.	
Preconditions	- The Consumer has previously generated and saved a personalised routine (UC1). - The dermatologist is verified and logged into the system. - The system has obtained Consumer consent to share the routine for professional review.	
Post Conditions, Success End Condition	- The dermatologist's feedback is recorded and stored in the system. - Feedback becomes available in the Consumer's profile. - The system logs the consultation and updates follow-up schedules.	
Long DESCRIPTION of Scenario	ThePerfectMatch system identifies a pending routine that requires expert review and transmits the related data to the dermatologist's dashboard. The dermatologist accesses the data through the professional portal, reviews the Consumer's profile and routine, and provides professional comments or updated product recommendations. Once feedback is submitted, the system validates and stores it securely, notifies the Consumer of its availability, and records the consultation for continuity and future improvement tracking.	
Main Flow		
Step	**Action**	**Alternate**
n.2.1	The system identifies a Consumer routine awaiting dermatologist review.	
n.2.2	The system transmits the relevant routine and Consumer data to the dermatologist portal.	
n.2.3	The dermatologist logs into the professional interface.	

n.2.4	The dermatologist accesses their assigned Consumer review dashboard.	
n.2.5	The dermatologist opens the Consumer's data package (routine, selfies, quiz results).	A.F. 2.5
n.2.6	The system verifies data integrity and completeness.	E.F. 2.6
n.2.7	The dermatologist analyses the provided data and identifies issues or improvements.	
n.2.8	The dermatologist adds written comments, suggested changes, or new product recommendations.	A.F. 2.8
n.2.9	The dermatologist submits the completed feedback through the portal.	E.F. 2.9
n.2.10	T he system validates and stores the dermatologist's feedback securely in the database.	
n.2.11	The system generates and queues a notification to inform the Consumer of new expert feedback.	E.F. 2.11
n.2.12	The consultation record is logged for analytics and potential follow-up scheduling.	A.F. 2.12
n.2.13	End Use Case.	
EXCEPTIONS or ERROR Flow Description		
Step	**Branching Action** < Exception number m of Use Case n>	**Alternate**
E.F. 2.6.1	Data package incomplete.	
E.F. 2.6.2	System requests retransmission.	
E.F. 2.9.1	Feedback submission fails.	
E.F. 2.9.2	Dermatologist prompted to retry.	
E.F. 2.9.3	Draft is automatically saved.	
E.F. 2.11.1	Notification delivery fails.	
E.F. 2.11.2	System retries.	
E.F. 2.11.3	System logs event.	
ALTERNATIVE or VARIATION Flow Description		
Step	**Branching Action**	**Alternate**
A.F. 2.5.1.	The dermatologist requests additional data.	
A.F. 2.5.2	The system generates a pending data request for the Consumer.	
A.F. 2.8.1	The dermatologist saves draft feedback for later completition.	
A.F. 2.8.2	The system flags the case as "In Progress".	
A.F. 2.12.1	The dermatologist schedules a follow-up session.	
A.F. 2.12.2	System updates the consultation calendar.	

RELATED INFORMATION	Use Case: 2	Get Expert Feedback
Priority:	High	
Product: Performance	Dermatologist dashboard must load assigned reviews within 3 seconds.	
Product: Reliability	Secure data transfer (99.9% uptime [3]).	

External: Legislation	All Consumer data must comply with GDPR [4] and medical data confidentiality requirements.
External: Ethical	Dermatologists must follow verifies, unbiased professional practice guidelines.
Channels to actors	Dermatologist interacts through the professional mobile/web portal only.
OPEN ISSUES	1. Define dermatologist verification and onboarding process. 2. Implement secure chat or video consultation feature.
Due Date	Feb 2026
...any other management information...	

2.2.3 Narrative for Use Case 3 – AR Try-On

USE CASE	Use Case: 3	AR Try-On
Brief Description of Goal in Context	The Consumer wants to virtually try on makeup or hair, nail, grooming products, using AR (e.g. preview foundation shades, or visualise skincare effects over time). The system uses the device camera and AI-generated overlays to provide realistic previews.	
Preconditions	- The Consumer has generated or selected a routine or product. - The device supports AR functionality. - The app has camera permissions.	
Post Conditions, Success End Condition	- AR preview is displayed successfully. - Consumer can save or apply products to their routine.	
Long DESCRIPTION of Scenario	The Consumer selects a product or routine to try in AR. The system activates the camera and overlays the chosen product on the Consumer's live image or selfie. AI adjusts the overlay based on facial features, skin tone, and lighting. The Consumer can adjust intensity or save a snapshot.	

Main Flow

Step	Action	Alternate
n.3.1	The Consumer opens ThePerfectMatch app.	
n.3.2	The Consumer selects "AR Try-On".	
n.3.3	The system requests camera access.	E.F. 3.3
n.3.4	The Consumer grants camera permission.	A.F. 3.4
n.3.5	The Consumer selects product or routine to preview.	
n.3.6	The system overlays the product/routine on live image.	E.F. 3.6
n.3.7	The Consumer adjusts intensity or shade.	
n.3.8	The Consumer saves AR preview.	
n.3.9	End Use Case.	

EXCEPTIONS or ERROR Flow Description

Step	Branching Action < Exception number m of Use Case n>	Alternate
E.F. 3.3.1	Camera access denied.	
E.F. 3.3.2	System prompt Consumer to enable.	

E.F. 3.6.1	AR overlay fails.	
E.F. 3.6.2	System retries.	
E.F. 3.6.3	System suggests using static image.	
ALTERNATIVE or VARIATION Flow Description		
Step	**Branching Action**	**Alternate**
A.F. 3.4.1	The Consumer denies camera access.	
A.F. 3.4.2	The system cancels AR Try-On.	
A.F. 3.4.3	The system returns to main dashboard.	
A.F. 3.4.4	Go to Main Flow 3.1	

RELATED INFORMATION	Use Case: 3	AR Try-On
Priority:	High – longer development time.	
Product: Performance	AR rendering under 2 seconds.	
Product: Reliability	Works on 95% of supported devices.	
External: Legislation	GDPR compliance for photos [4].	
Channels to actors	Consumer interacts through mobile app UI and device camera interface.	
OPEN ISSUES	1. Ensure AR performs accordingly on all devices (Android, iOS, etc.).	
Due Date	Feb 2026	
...any other management information...		

2.2.4 Narrative for Use Case 4 – Process Consumer Data

USE CASE	Use Case: 4	Process Consumer Data
Brief Description of Goal in Context	The system analyses Consumer inputs (selfies, quiz answers, sensor data) to extract characteristics like skin type, hair condition, preferences, and potential sensitivities to generate structured data for routines or dermatologist review.	
Preconditions	- The Consumer provides valid input data. - The system is connected to the database.	
Post Conditions, Success End Condition	- Structured Consumer data is stored and available for routine generation or professional feedback.	
Long DESCRIPTION of Scenario	When the Consumer submits input, the system automatically processes it using trained models. Skin and hair features are identified (e.g. dryness, oiliness, acne, scalp health), preferences are extracted, and structured Consumer data is stored in the profile for tasks like AR previews, routine generation, or professional consultation.	
Main Flow		
Step	**Action**	**Alternate**
n.4.1	The Consumer submits input (selfie, quiz, sensor).	
n.4.2	The system verifies data completeness.	E.F. 4.2
n.4.3	The system extracts skin/hair features.	

n.4.4	The system matches attributes with known conditions/preferences.	
n.4.5	The processed data is stored in the Consumer profile.	
n.4.6	End Use Case.	
EXCEPTIONS or ERROR Flow Description		
Step	**Branching Action** < Exception number m of Use Case n>	**Alternate**
E.F. 4.2.1	Input incomplete.	
E.F. 4.2.2	System prompts Consumer to resubmit.	
ALTERNATIVE or VARIATION Flow Description		
Step	**Branching Action** No alternative flows.	**Alternate**

RELATED INFORMATION	Use Case: 4	Process Consumer Data
Priority:	High	
Product: Performance	Processing under 5s.	
Product: Reliability	99.9% uptime [3].	
External: Legislation	All Consumer data must comply with GDPR [4].	
External: Ethical	Must comply with medical data confidentiality.	
Channels to actors	Consumer interacts through mobile app UI. Sensors connect via Bluetooth. The system interacts with internal databases.	
OPEN ISSUES	1. Evaluate data storage encryption methods. 2. Train the system to accomplish requested tasks accordingly.	
Due Date	Dec 2025	
...any other management information...		

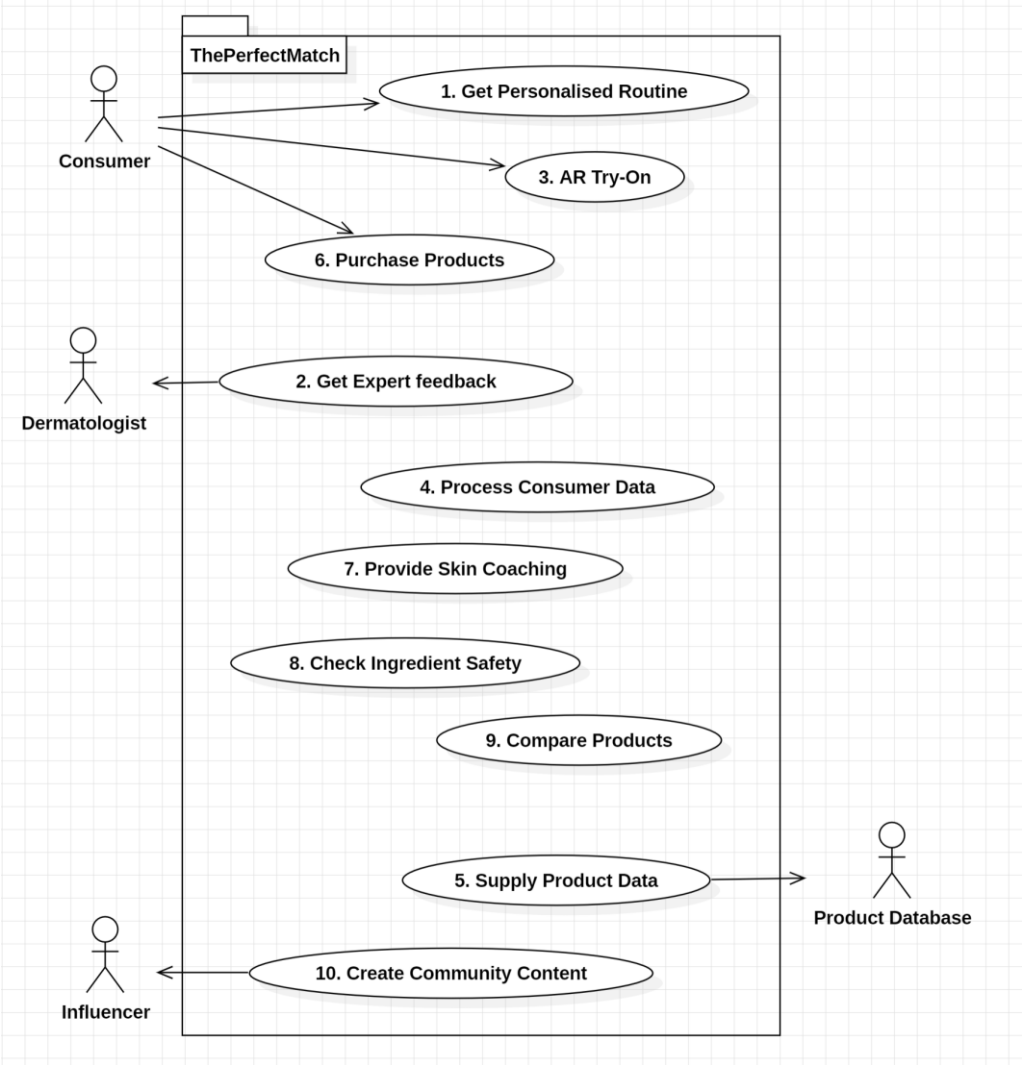
2.2.5 Narrative for Use Case 5 – Supply Product Data

USE CASE	Use Case: 5	Supply Product Data
Brief Description of Goal in Context	The Product Database provides verified information on beauty and self-care products for AI and Consumer queries, including ingredients, safety, price, eco-friendliness, and certifications.	
Preconditions	- Database connection active. - API or internal dataset synced and verified.	
Post Conditions, Success End Condition	- Product data is correctly retrieved and displayed. - AI and Consumer queries are accurately satisfied.	
Long DESCRIPTION of Scenario	When requested by the AI Engine or Consumer, the product database searches for matching products, filters results by safety, skin/hair compatibility, and availability, then returns structured information. This ensures that recommendations are accurate, safe, and aligned with Consumer preferences.	
Main Flow		
Step	**Action**	**Alternate**
n.5.1	The AI Engine sends product query.	E.F. 5.1
n.5.2	The Product Database searches for relevant products.	E.F. 5.2

n.5.3	The system filters the products by safety, skin compatibility, and availability.	
n.5.4	The filtered data is transmitted to the AI Engine/Consumer.	
n.5.5	AI integrates data into final output.	
n.5.6	End Use Case.	
EXCEPTIONS or ERROR Flow Description		
Step	**Branching Action** < Exception number m of Use Case n>	**Alternate**
E.F. 5.1.1	Database query fails.	
E.F. 5.1.2	System retries.	
E.F. 5.1.3	System notifies Consumer.	
E.F. 5.2.1	No matching product found.	
E.F. 5.2.2	System suggests broadening search.	
ALTERNATIVE or VARIATION Flow Description		
Step	**Branching Action** No alternative flows.	**Alternate**

RELATED INFORMATION	Use Case: 5	Supply Product Data
Priority:	High – essential for correct functionality of the app.	
Product: Performance	Search response under 3s.	
Product: Reliability	Database uptime 99.9% [3].	
External: Legislation	Complied with verified product sourcing.	
External: Ethical	Exclude harmful or unverified items.	
Channels to actors	Product Database interacts via API endpoints with the AI Engine and app back-end.	
OPEN ISSUES	1. Confirm external product API partners. 2. Implement automated data validation.	
Due Date	Dec 2025	
...any other management information...		

3. Third Iteration
3.1 Use Case Diagram



3.2 Use Case Narratives for Iteration 3
3.2.1 Narrative for Use Case 1 – Get Personalised Routine
Same as Iteration 2.

3.2.2 Narrative for Use Case 2 – Get Expert Feedback
Same as Iteration 2.

3.2.3 Narrative for Use Case 3 – AR Try-On
Same as Iteration 2.

3.2.4 Narrative for Use Case 4 – Process Consumer Data
Same as Iteration 2.

3.2.5 Narrative for Use Case 5 – Supply Product Data
Same as Iteration 2.

3.2.6 Narrative for Use Case 6 – Purchase Products

USE CASE	Use Case: 6	Purchase Products
Brief Description of Goal in Context	The Consumer purchases products directly from within ThePerfectMatch app using verified retailer links and secure payment integration.	
Preconditions	- The Consumer is logged in. - The selected product exists in the verified product database. - Secure payment gateway and internet connection are active.	
Post Conditions, Success End Condition	- Order is placed successfully. - Receipt is saved in the Consumer's purchase history.	
Long DESCRIPTION of Scenario	The Consumer selects a product recommended in their routine or found through browsing. The system retrieves the verified retailer link and opens the in-app store interface. The Consumer reviews details, adds the item to their cart, and confirms the order. Payment is securely processed, and the app displays confirmation with estimated delivery time. A copy of the receipt is stored in the Consumer's account for tracking.	

Main Flow		
Step	**Action**	**Alternate**
n.6.1	The Consumer selects a product for purchase.	
n.6.2	The system opens the in-app purchase page.	
n.6.3	The Consumer adds the product to their cart.	
n.6.4	The system verifies product availability.	E.F. 6.4
n.6.5	The Consumer confirms order.	A.F. 6.5
n.6.6	The Consumer inserts payment details.	
n.6.7	The system processes the payment securely.	E.F. 6.7
n.6.8	Receipt is generated.	
n.6.9	End Use Case.	
EXCEPTIONS or ERROR Flow Description		
Step	**Branching Action** < Exception number m of Use Case n>	**Alternate**
E.F. 6.4.1	Product out of stock.	
E.F. 6.4.2	System suggests similar alternatives.	
E.F. 6.4.3	Go to Main Flow 6.1	
E.F. 6.7.1	Payment failure.	
E.F. 6.7.2	System prompts the Consumer to retry.	
ALTERNATIVE or VARIATION Flow Description		
Step	**Branching Action**	**Alternate**
A.F. 6.5.1	The Consumer cancels order before payment.	
A.F. 6.5.2	System returns to product page.	

RELATED INFORMATION	Use Case: 6	Purchase Products
Priority:	High	
Product: Performance	Transaction completed within 5s of confirmation.	

Product: Reliability	Secure PCI-DSS compliant payment with 99.9% uptime [3].
External: Legislation	Complied with GDPR and eCommerce regulations.
Channels to actors	Consumer interacts through mobile app UI.
OPEN ISSUES	1. Implement multiple payment methods. 2. Add refund management system.
Due Date	Apr 2026
...any other management information...	

3.2.7 Narrative for Use Case 7 – Provide Skin Coaching

USE CASE	Use Case: 7	Provide Skin Coaching
Brief Description of Goal in Context	The system delivers personalised daily skin coaching by analysing Consumer data, environmental factors, and progress trends to recommend improvements and send actionable advice.	
Preconditions	- The system is connected to Consumer profile data. - The Consumer has enabled skin coaching notifications.	
Post Conditions, Success End Condition	- Personalised advice generated and displayed to the Consumer.	
Long DESCRIPTION of Scenario	ThePerfectMatch system analyses the Consumer's recent skincare data, such as product usage, environmental conditions, and previous dermatologist feedback. It compares the data to past records and identifies any changes in the Consumer's condition or routine consistency. Based on the analysis, the system generates a personalised message, such as "Use lighter moisturiser today" or "Avoid sun exposure." The message is sent as a notification and displayed on the Consumer's dashboard.	

Main Flow

Step	Action	Alternate
n.7.1	The system retrieves recent Consumer data.	E.F. 7.1
n.7.2	The system analyses data for issues.	
n.7.3	The system identifies potential improvements.	
n.7.4	The system generates personalised coaching advice.	E.F. 7.4
n.7.5	The system sends the advice to the Consumer's notification center.	A.F. 7.5
n.7.6	End Use Case.	

EXCEPTIONS or ERROR Flow Description

Step	Branching Action < Exception number m of Use Case n>	Alternate
E.F. 7.1.1	Missing Consumer data.	
E.F. 7.1.2	System logs missing data.	
E.F. 7.1.3	System notifies Consumer.	
E.F. 7.1.4	System prompts Consumer to input data.	
E.F. 7.4.1	Advice generation error.	
E.F. 7.4.2	System retries after next sync.	

ALTERNATIVE or VARIATION Flow Description

Step	Branching Action	Alternate

A.F. 7.5.1	The Consumer disables notifications.	
A.F. 7.5.2	Advice stored only in Consumer's dashboard.	

RELATED INFORMATION	Use Case: 7	Provide Skin Coaching
Priority:	High	
Product: Performance	Advice generated within 5 seconds.	
Product: Reliability	System available 99.9% of the time [3].	
External: Ethical	Disclaimer: not a medical substitute.	
Channels to actors	System sends output to the Consumer dashboard and notification centre.	
OPEN ISSUES	1. Add feedback scoring to measure advice helpfulness. 2. Link to Dermatologist mode for critical advice.	
Due Date	Apr 2026	
...any other management information...		

3.2.8 Narrative for Use Case 8 – Check Ingredient Safety

USE CASE	Use Case: 8	Check Ingredient Safety
Brief Description of Goal in Context	The system analyses the safety and compatibility of ingredients in a given product and returns an allergen report to the system for display.	
Preconditions	- Product Database connection is active. - Valid ingredient list or product barcode available.	
Post Conditions, Success End Condition	- Ingredient safety analysis completed. - Report returned and displayed to the Consumer.	
Long DESCRIPTION of Scenario	The system receives a list of ingredients from a product the Consumer scanned or entered manually. It matches the names to verified database entries, and checks for toxic, allergenic, or restricted compounds. A safety summary is generated, showing risk levels and suggested alternatives. This report is then stored for the Consumer's future reference.	

Main Flow		
Step	**Action**	**Alternate**
n.8.1	The system receives the ingredient data.	E.F. 8.1
n.8.2	The system verifies ingredient names.	A.F. 8.2
n.8.3	The system cross-references data with the product database.	E.F. 8.3
n.8.4	The system identifies unsafe ingredients.	A.F. 8.4
n.8.5	The system generates a safety summary.	
n.8.6	The system returns the report to the app.	
n.8.7	The system saves the report for future use.	
n.8.8	End Use Case.	

EXCEPTIONS or ERROR Flow Description		
Step	**Branching Action** < Exception number m of Use Case n>	**Alternate**
E.F. 8.1.1	Incomplete data.	
E.F. 8.1.2	System requests resubmission.	

E.F. 8.3.1	Database query fails.	
E.F. 8.3.2	System retries searching.	
ALTERNATIVE or VARIATION Flow Description		
Step	**Branching Action**	**Alternate**
A.F. 8.2.1	Ingredient not recognised.	
A.F. 8.2.2	The system logs the event.	
A.F. 8.2.3	The system prompts the Consumer to verify data correctness.	

RELATED INFORMATION	Use Case: 8	Check Ingredient Safety
Priority:	High.	
Product: Performance	Analysis completed in under 5 seconds.	
External: Legislation	Must comply with EU Cosmetic Safety Regulations and GDPR.	
Channels to actors	System interacts with internal product safety database.	
OPEN ISSUES	1. Expand database with international brands 2. Add visual allergen badges.	
Due Date	Apr 2026	
...any other management information...		

3.2.9 Narrative for Use Case 9 – Compare Products

USE CASE	Use Case: 9	Compare Products
Brief Description of Goal in Context	The system compares two or more products based on Consumer-defined attributes (price, brand, composition, and eco-rating) and produces a ranked comparison report.	
Preconditions	- Product information exists in the verified database. - Consumer selected two or more products to compare.	
Post Conditions, Success End Condition	- Comparison table and rankings generated and displayed.	
Long DESCRIPTION of Scenario	The system retrieves product details from the product database, such as ingredients, brand, sustainability rating, and average Consumer score. Using this data, it performs a multi-criteria comparison and generates a summary showing the best match per attribute. Results are returned to the app for display to the Consumer.	
Main Flow		
Step	**Action**	**Alternate**
n.9.1	The system receives product IDs from the app.	
n.9.2	The system retrieves product attributes from the database.	E.F. 9.2
n.9.3	The system analyses the selected attributes.	A.F. 9.3
n.9.4	The system compares the attributes.	
n.9.5	A ranked summary table is generated.	
n.9.6	The results are transmitted back to the system.	
n.9.7	End Use Case.	

EXCEPTIONS or ERROR Flow Description		
Step	**Branching Action** < Exception number m of Use Case n>	**Alternate**
E.F. 9.2.1	Missing product data.	
E.F. 9.2.2	System flags incomplete comparison.	
E.F. 9.2.1	No matching product found.	
E.F. 9.2.2	System suggests broadening search.	
ALTERNATIVE or VARIATION Flow Description		
Step	**Branching Action**	**Alternate**
A.F. 9.3.1	Consumer-defined weights applied to comparison (e.g. eco > price).	

RELATED INFORMATION	Use Case: 9	Compare Products
Priority:	Medium	
Product: Performance	Comparison generated in under 3 seconds.	
Product: Reliability	Database uptime 99.9% [3].	
External: Ethical	Avoid commercial bias in ranking.	
Channels to actors	System communicates with Product Database API.	
OPEN ISSUES	1. Add brand reputation scoring. 2. Include dermatologist verification.	
Due Date	May 2026	
...any other management information...		

3.2.10 Narrative for Use Case 10 – Create Community Content

USE CASE	Use Case: 10	Create Community Content
Brief Description of Goal in Context	The influencer shares content such as skincare routines, product reviews, or educational posts with the app's Consumer community to increase engagement and knowledge exchange.	
Preconditions	- The influencer has a verified account.	
Post Conditions, Success End Condition	- Content published to the community feed. - Engagement metrics updated.	
Long DESCRIPTION of Scenario	The influencer logs in to the PerfectMatch app and accesses the community dashboard. They select "Create Content," upload media or write a post, and tag relevant products or routines. The system scans the submission for inappropriate content, then publishes it once validated. The post becomes visible to Consumers who can like or comment.	

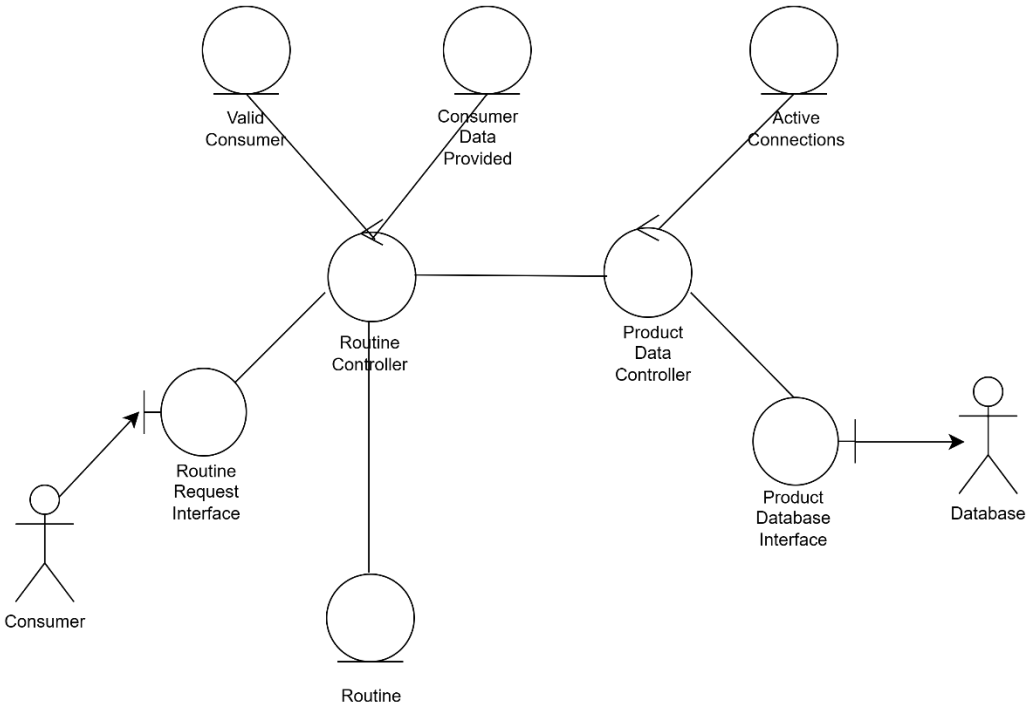
Main Flow		
Step	**Action**	**Alternate**
n.10.1	The influencer logs into the app.	
n.10.2	The system verifies influencer status.	
n.10.3	The influencer selects "Create Community Content."	
n.10.4	The influencer uploads text, images, or videos.	A.F. 10.4
n.10.5	The system reviews the content for compliance.	E.F. 10.5
n.10.6	The post is published to the community feed.	

n.10.7	Engagement metrics updated.	
n.10.8	End Use Case.	
EXCEPTIONS or ERROR Flow Description		
Step	**Branching Action** < Exception number m of Use Case n>	**Alternate**
E.F. 10.5.1	Inappropriate content detected.	
E.F. 10.5.2	System rejects submission.	
ALTERNATIVE or VARIATION Flow Description		
Step	**Branching Action**	**Alternate**
A.F. 10.4.1	Influencer saves post as draft.	

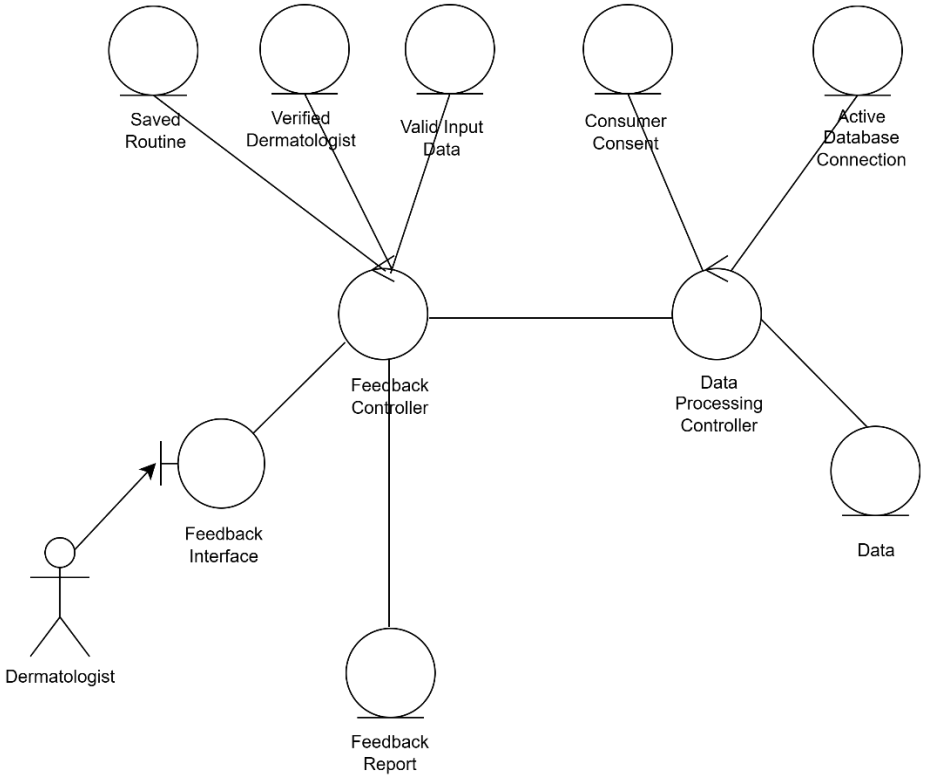
RELATED INFORMATION	Use Case: 10	Create Community Content
Priority:	Medium	
Product: Performance	Upload and publication in under 5 seconds.	
Product: Reliability	99.9% uptime for community services [3].	
External: Ethical	Must comply with digital content guidelines (no misleading claims).	
Channels to actors	Influencer interacts through mobile creator portal.	
OPEN ISSUES	1. Introduce comment moderation system. 2. Add collaboration tagging for other influencers.	
Due Date	May 2026	
...any other management information...		

4. Analysis Classes

4.1 Pair: Get Personalised Routine & Supply Product Data

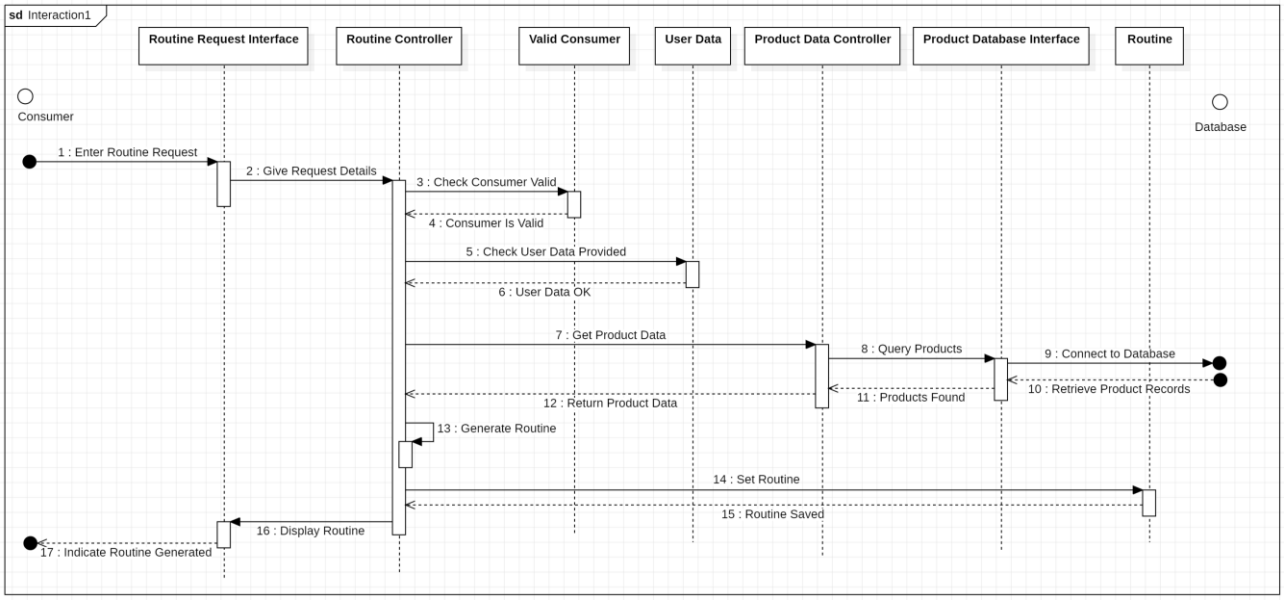


4.2 Pair: Get Expert Feedback & Process Consumer Data

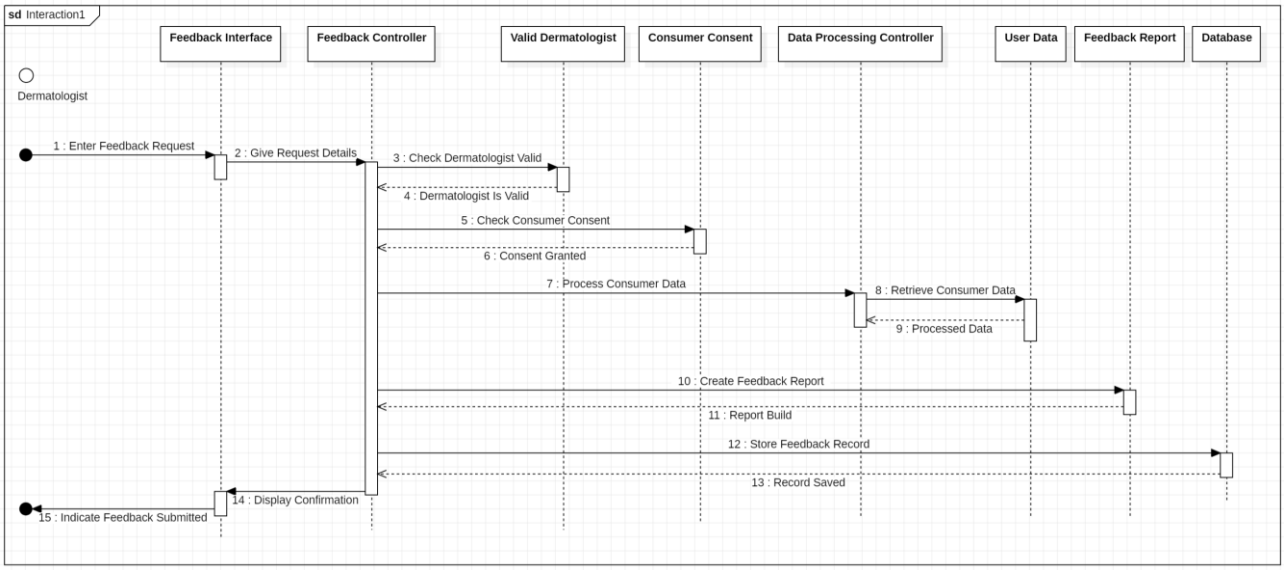


5. Sequence Diagrams

5.1 Pair: Get Personalised Routine & Supply Product Data



5.2 Pair: Get Expert Feedback & Process Consumer Data



6. References

[1] “Virtual Try-On Technology Guide”, 2025, Andrew Makarov, Mobidev, https://mobidev.biz/blog/augmented-reality-virtual-try-on-technology-for-ecommerce
[2] “Optimize your app’s size and stay within Google Play app size limits”, 2025, Google Help, https://support.google.com/googleplay/android-developer/answer/9859372?hl=en
[3] “The Rule of Nines: Why You Need 99.999% Availability in an Always-On World”, Penguin Solutions, https://www.penguinsolutions.com/en-us/resources/blog/rule-nines-availability-always-on-world#:~:text=What%20does%2099%25%20uptime%20mean,of%20unplanned%20downtime%20per%20year.
[4] “GDPR Compliance for Apps: A 2025 Guide”, 2025, Ana Mishova, GDPR Local, https://gdprlocal.com/gdpr-compliance-for-apps/